



P3HT:Borophene-ZnO P-N Junction Based Thermoelectric Generator

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Received: 27 August 2025 / Accepted: 24 September 2025 / Published online: 13 October 2025
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Abstract

This study presents the development and characterization of P3HT: Borophene-ZnO p-n junction-based organic thermoelectric generators (OTEGs), demonstrating superior performance compared to P3HT-ZnO systems. The incorporation of borophene significantly enhances electrical conductivity (reaching $\sim 12 \times 10^4$ S/cm) while maintaining favorable Seebeck coefficients (~ 100 – 200 μ V/K), particularly in the 400–500 K range, due to borophene's 2D conductive network that bridges the organic-inorganic interface. Structural analyses confirm the crystalline quality of borophene and its effective integration with P3HT. At the same time, thermoelectric tests show that the ternary composite achieves higher power factors than the binary P3HT-ZnO system, making it a promising candidate for mid-temperature waste heat recovery applications. The work advances organic thermoelectrics by optimizing the synergy between conductive 2D materials (borophene), polymers (P3HT), and metal oxides (ZnO) through innovative solution processing and junction engineering.

Keywords Energy, energy generation · thermoelectric generator · borophene · poli(3-heksiltiofen), zinc oxide

Introduction

With the market for TEGs becoming increasingly competitive, it has become imperative for researchers to innovate in ways that enable TEGs to compete effectively on the global stage, offering improved performance, lower costs, and greater reliability.^{1,2}

As semiconductor polymers have high electrical conductivity (σ), high Seebeck coefficient (S), and low thermal conductivity, they have been used in TEG designs.^{3–5} Poly(3-hexylthiophene-2,5-diyl) (P3HT), a p-type semiconductor with improved electrical conductivity, is also being investigated in TEG architectures among semiconductor polymers. Interesting studies have been carried out on P3HT-based TEGs.

Various nanocomposites containing P3HT polymer and carbon nanotubes (CNTs) have been prepared,^{6–9} and TEGs with a power factor (PF) of up to $572 \mu\text{W m}^{-1} \text{K}^{-2}$ have been obtained.⁹ These studies have shown that the thermoelectric performance of the polymer can be significantly enhanced when the P3HT polymer is combined with CNTs into a nanocomposite. In addition, P3HT-based TEGs integrated into textiles have been prepared.^{10,11} The obtained performance results indicate that P3HT composites are promising for wearable and flexible energy harvesting applications. Additionally, molecular and additive doping studies were conducted to increase the conductivity and Seebeck coefficients. PF values of ~ 1 – $8 \mu\text{W K}^{-2}$ were achieved using F4TCNQ-doped P3HT thin films.^{12,13} By improving the performance using AuCl_3 doping, electrical conductivity of 207 S cm^{-1} and power factor of $110 \mu\text{W m}^{-1} \text{K}^{-2}$ with a flexible OTEG output voltage of 7.96 mV was achieved.¹⁴ A power factor of $464 \mu\text{W m}^{-1} \text{K}^{-2}$ was achieved using Pt-based π -ligand complexes.¹⁵ All these findings indicate that chemical doping of P3HT polymer is one of the most powerful strategies to optimize thermoelectric efficiency. These studies demonstrate significant progress in optimizing the Seebeck coefficient, electrical conductivity, and power factor of P3HT polymers. In this context, borophene, a boron atom monolayer with rich polymorphic structures, stands out as a promising candidate for enhanced charge transport and

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improved thermoelectric performance when incorporated into polymer-based systems due to its high carrier mobility, anisotropic thermal transport, and strong chemical reactivity.

Besides these, Zinc Oxide (ZnO), with its wide band gap (3.37 eV), high exciton-binding energy (60 meV), and excellent thermal stability, is a promising n-type thermoelectric (TE) material for high-temperature applications. While undoped ZnO exhibits a high Seebeck coefficient ($\sim 478 \mu\text{V/K}$) but low power factor ($\text{PF} \sim 0.75 \times 10^{-4} \text{ W m}^{-1} \text{ K}^{-2}$ at room temperature), doping strategies significantly enhance its TE performance - Al doping improves electrical conductivity, with $\text{Zn}_{0.96}\text{Al}_{0.02}\text{Ga}_{0.02}\text{O}$ achieving a record PF of $23.9 \times 10^{-4} \text{ W m}^{-1} \text{ K}^{-2}$ at $\sim 1147 \text{ K}$, while Dy doping ($\text{Zn}_{0.995}\text{Dy}_{0.005}\text{O}$) boosts PF to $4.46 \times 10^{-4} \text{ W m}^{-1} \text{ K}^{-2}$ at 1100 K , 56 times higher than undoped ZnO.^{16–24}

Inspired by all these studies, in this study, the effect of P3HT and borophene nanocomposite on TEG performance when used in TEG architecture was systematically investigated.

Experimental

Preparation and Characterization of the P3HT: Borophene

Boron microparticles (1 μm particle size) were purchased from Nanografi Company. N, N-Dimethylformamide (DMF) was purchased from the Company. P3HT and ZnO were purchased from Sigma Aldrich. To prepare borophene, 100 mg of boron microparticles was sonicated in 100 ml of DMF at 200 W for 4 h in a nitrogen (N_2) flow-controlled cabinet to prevent oxidation during the physical exfoliation process. The exfoliated boron solution was centrifuged at 5000 rpm and 12,000 rpm for 15 min, respectively. The resulting borophene was carefully collected and dried in the vacuum chamber at 50°C for 4 h. A 1 g P3HT solution was prepared by

dissolving 1 g of P3HT in 50 mL of chloroform. To obtain P3HT: Borophene, 20 mg of borophene was added to a 50 mL P3HT solution, and the mixture was sonicated at 50 W for 15 min. Additionally, a ZnO solution was prepared by dissolving 200 mg of ZnO powder into 50 mL of deionized water at room temperature.

Structural, morphological, and chemical analyses of the prepared materials were conducted using various techniques. High-Resolution Transmission Electron Microscopy (HRTEM) (JEOL JEM-ARM200CFEG UHR-TEM), Selected Area Electron Diffraction (SAED), x-ray Diffraction (XRD) (PAN analytical Xpert-Pro), x-ray Diffraction (XRD) with Cu-K α radiation (PAN analytical Xpert-Pro) ($\lambda = 1.5406 \text{ \AA}$), and FTIR (Perkin Elmer, Spectrum Two, frequencies between 4000 and 500 cm^{-1} , 4 cm^{-1} and 8 scan resolution) techniques.

Preparation and Thermoelectric Tests of the P3HT: Borophene Junction and P3HT: Borophene-ZnO P-N Junction OTEGs

The TEG module substrate was prepared by creating 10 p-n electrode channels using a 3D printer. Copper wire was used to establish the connections between the p-n electrodes. The electrodes were prepared within the TEG module substrate (Fig. 1). According to the literature, borophene is a p-type semiconductor material.²⁵ Therefore, both the plain and borophene-modified forms of the P3HT p-type semiconductor polymer were used as p-type electrodes. ZnO was used as the n-type electrode in all studies, as it is a solution-processable material that is both cost-effective and demonstrates optimal n-type electrode performance.^{19,20} This allowed for a comparative evaluation of the p-type electrodes.

P3HT, a p-type semiconductor polymer in its pure form combined with borophene, was used to prepare p-type electrodes, while a ZnO solution was used to prepare n-type electrodes. P3HT, P3HT: Borophene and ZnO were filled in

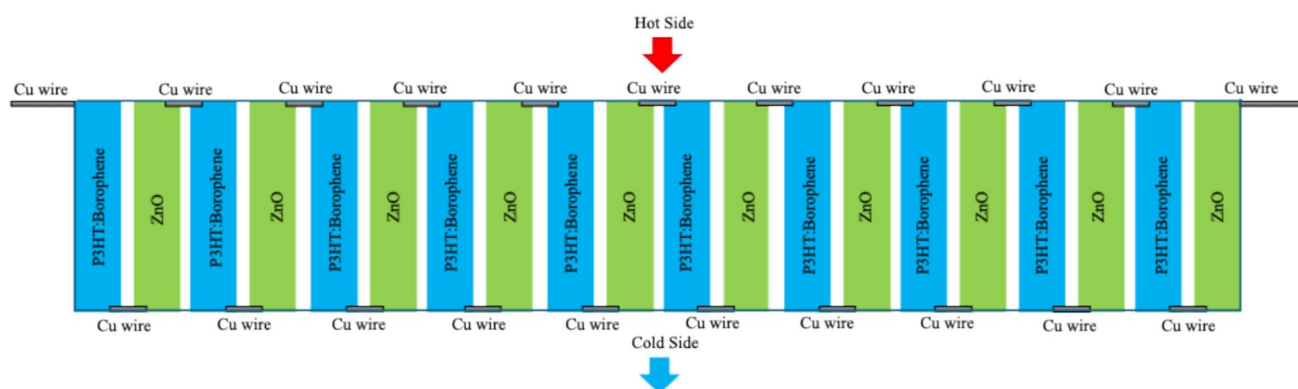


Fig. 1 The electrodes within the prepared P3HT: Borophene-ZnO P-N junction based OTEG.

the channels of the TEG module by dripping, forming P3HT, P3HT: Borophene, and ZnO electrodes within the module. P3HT-ZnO junction TEGs and P3HT: Borophene-ZnO junction TEGs were fabricated with different electrode thicknesses (5, 10, and 20 μm). Subsequently, the TEG module containing P3HT, P3HT: Borophene, and ZnO was placed in a vacuum chamber and dried at 90 °C for 5 minutes. Since Copper (Cu) wires were pre-installed in the module to establish contact between the p-type and n-type channels, forming a p-n junction, the module was then placed in the test system and evaluated.

The prepared TEG module was set up so that half of it was on a metal block with temperature control, and the other half remained at room temperature. To create a temperature gradient throughout the module, the metal block was heated from 25 to 150 °C. Using a heater, the temperature on the hot side-which corresponded to the heated metal block-was changed from 25 to 150 °C. The temperatures at the interface of the hot and cold sides of the TEG were directly measured using a thermocouple to a thermally conductive material. The thermovoltage between the hot and cold sides was measured using two aluminum contact tips attached to the module with the GW Instek GDM83-51 device. The test system software was developed in MATLAB software.

Results and Discussion

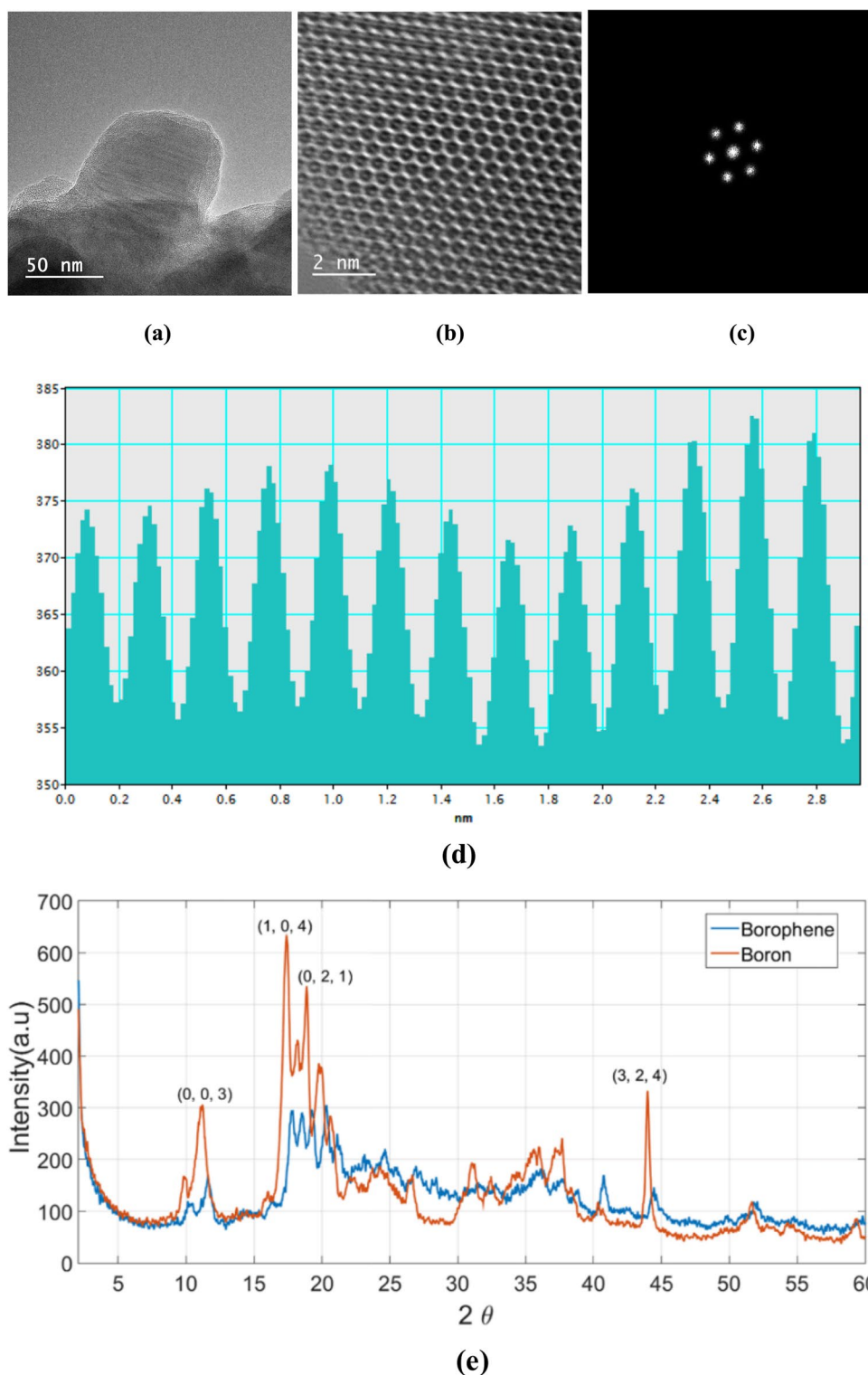
Characterization Results of the Borophene and P3HT: Borophene and ZnO

In this study, the crystal phase, planar orientation, regularity of borophene and the quality of the physical exfoliation process were revealed with the data obtained from HRTEM, SAED and XRD analysis results (Fig. 2). As seen from the HRTEM images of borophene nanosheets (Fig. 2a and b), borophene nanosheets formed semi-transparent, thin, and large surface structures. There were slight curls on some edges of the nanosheets. This situation shows that borophene has high flexibility. The fact that the structures were layerless and thin indicates the success of the physical exfoliation process. The high-resolution HRTEM image provided a clear observation of the atomic arrangement of borophene, and it was observed that the obtained borophene was in the beta phase and had a planar lattice structure. As seen from the SAED image of borophene nanosheets (Fig. 2c), the circular ring patterns in the SAED image indicate that borophene has a crystalline nature corresponding to the (003), (021), and (324) planes. These planes were consistent with the crystal planes reported in the literature for beta phase borophene. The prominence of the circular rings indicates that these crystal structures have close orientations and that borophene has high crystallinity. The (021) plane formed a particularly

strong diffraction ring, which was densely represented, and it was understood that borophene grew in this orientation. As seen from the profile pattern of borophene nanosheets (Fig. 2d), because of the measurement of the lattice fringe spacings, values of approximately 0.23 nm and 0.38 nm were obtained, corresponding to the (021) and (003) planes reported for beta phase borophene in the literature. In particular, the fact that these planes can be separated very clearly reveals that the crystal contains single crystal-like areas and was regular. It was thought that physical exfoliation not only provides surface peeling but also provides layering at the atomic level. The fact that the nanosheets do not overlap topographically and that side-by-side planes can be followed indicates that the physical exfoliation process performed in this study was carried out with high efficiency. This result was consistent with the XRD analysis result. As seen from the XRD pattern of borophene nanosheets (Fig. 2e), the peaks correspond to the (003), (104), (021), and (324) planes, respectively, indicating that borophene was in the beta phase according to PDF31-0207 card.²⁶ The (003) plane ($\approx 8.6^\circ$) represents the layered structure of borophene along the c-axis. Although the peak height is relatively low, it is important that this plane maintained its existence after exfoliation. The dominance of the (021) plane ($\approx 19.6^\circ$) peak showed that borophene nanosheets mostly grew in this plane and the surface orientation occurred accordingly. This plane was also dominant in the SAED analysis. The high-angle (324) plane ($\approx 39.8^\circ$) indicates that deep planar reflections occur within the borophene crystal structure. Despite its low intensity, the observability of this situation was an indication of the high degree of crystallinity of borophene. The wide peaks indicate nanocrystal sizes. Calculations made according to the Scherrer equation showed that the average crystallite size was in the range of 5–10 nm and the crystal regions formed were compatible with 2D structures, and borophene was a 2D nanomaterial. Moreover, the calculated band gap energies of the prepared borophene are 0.94 eV for the indirect transition and 1.74 eV for the direct transition.²⁷

Furthermore, the FTIR plots of the P3HT, the prepared borophene, and the prepared P3HT: Borophene are shown in Fig. 3. The characteristic peaks of the prepared borophene were found at 3487.31 cm^{-1} (O–H stretching or N–H stretching vibrations), 2930.59 cm^{-1} (B–B bond), 2861.19 cm^{-1} (B–H bond vibration bands), 1653.73 cm^{-1} (C=O stretching), 1495.52 cm^{-1} (B–H bond vibrations), 1385.07 cm^{-1} (B–O bonding), 1254.48 cm^{-1} (B–O bonding), 1091.04 cm^{-1} (B–O–B vibrations), and 654.49 cm^{-1} (C–H), respectively. The fundamental bands of P3HT are observed in the FTIR graph as aliphatic and aromatic C–H stretching vibrations (thiophene ring and side chains) at 3000–2800 cm^{-1} ; conjugated C=C stretching (main skeleton of polymer) at 1500–1450 cm^{-1} ; and C–H bonding vibrations of thiophene ring at 820–800 cm^{-1} . Band shifts

Fig. 2 (a) HRTEM image, (b) High-resolution HRTEM image, (c) SAED image, (d) profile pattern, and (e) XRD pattern of the precursor boron and prepared borophene.



were observed in P3HT: Borophene. The π - π interactions or electron transfer between borophene and P3HT caused slight shifts in the C=C (1500 cm^{-1}) or C-H ($2800\text{--}3000\text{ cm}^{-1}$) bands. The C=C stretching band shifted to lower wavenumbers by $10\text{--}20\text{ cm}^{-1}$ (due to increased electron density). B-B ($\sim 1200\text{ cm}^{-1}$) and B-O ($\sim 1300\text{ cm}^{-1}$) bands belonging to

borophene were observed. The intensity of the C-H bands decreased due to the arrangement of the polymer chains or interaction with borophene. Electron delocalization occurred between the 2D structure of borophene and the conjugated chains of P3HT. The -OH or -B=O groups on borophene formed hydrogen bonds with the sulfur atoms of P3HT.

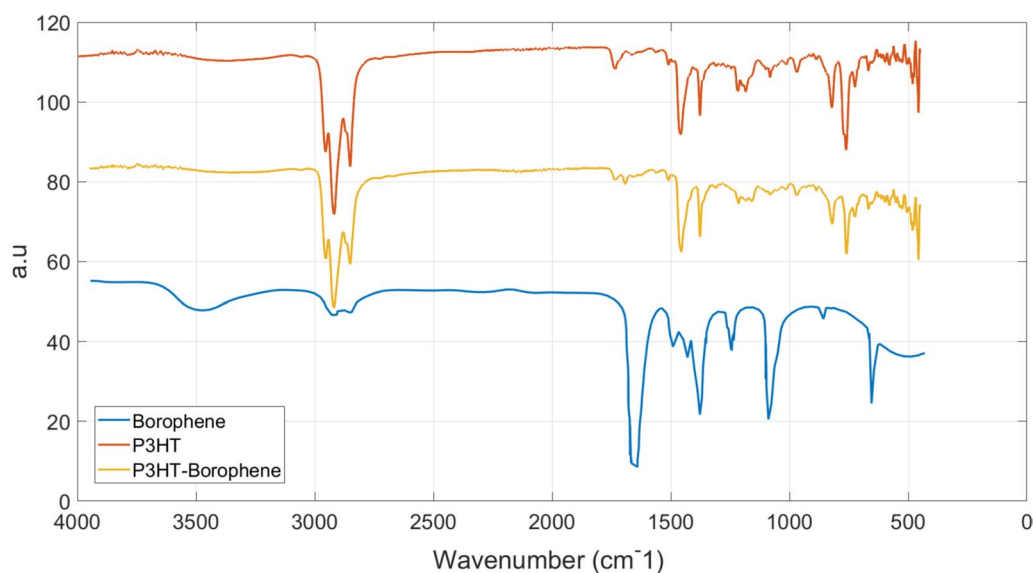


Fig. 3 FTIR plots of the P3HT, the prepared borophene, and the prepared P3HT: Borophene.

Thermoelectric Characteristics of the P3HT-ZnO P-N Junction and P3HT: Borophene-ZnO P-N Junction based OTEGs

The electrical conductivity-temperature relation and the Seebeck coefficient-temperature relation of the P3HT-ZnO and P3HT: Borophene-ZnO P-N junction-based OTEGs are shown in Fig. 4. The thermoelectric characteristics of a P3HT-ZnO composite, plotting electrical conductivity against temperature (300–500 K). While the exact conductivity values are not shown, the temperature range suggests that the material exhibits increasing conductivity with rising temperature. In typical semiconductor behavior, thermal energy promotes the mobility of charge carriers. Since ZnO is an n-type semiconductor and P3HT is a p-type semiconductor, it is thought that the conductivity of the composite increases when ZnO is added to the composite (Fig. 4a). The thermoelectric performance of the P3HT: Borophene-ZnO composite, showing the temperature-dependent (300–500 K) changes in electrical conductivity and Seebeck coefficient (S). The electrical conductivity increases significantly with temperature (from ~ 2 to $\sim 12 \times 10^4$ S/cm), demonstrating the conductive network provided by the Borophene-ZnO framework and enhanced charge carrier mobility in P3HT. Peak performance is measured around 400 K. The slight decrease in electrical conductivity at higher temperatures may be due to increased carrier scattering (Fig. 4b).

The comparative analysis of P3HT-ZnO and P3HT: Borophene-ZnO composites reveals significant performance differences. While both systems show temperature-dependent behavior (300–500 K), the Borophene-containing composite demonstrates superior electrical conductivity (reaching

12×10^4 S/cm versus unspecified but presumably lower values for P3HT-ZnO) and likely enhanced thermoelectric efficiency. The addition of Borophene creates a more robust conductive network, as evidenced by the higher conductivity values and potentially better-maintained Seebeck coefficients. Power factor of the P3HT: Borophene-ZnO P-N junction based OTEG was higher. Both materials follow the typical semiconductor trend of increasing conductivity with temperature. The P3HT-ZnO may indicate faster charge transport at higher temperatures, but this comes at the apparent cost of lower overall conductivity compared to the Borophene-enhanced version. The Borophene-containing composite appears particularly optimized for the 400–500 K range, where it maintains high conductivity while likely preserving reasonable Seebeck values, making it the more promising candidate for practical thermoelectric applications. These results make the P3HT: Borophene-ZnO composite suitable for applications requiring stable performance, such as thermoelectric energy harvesting applications over a wide temperature range.

In the measurements, the open-circuit voltage (V_{oc}) increased almost linearly with ΔT , indicating stable charge carrier transport within P3HT due to borophene. This nearly linear behavior indicates that undesirable thermal and electrical barriers at the P3HT-borophene interface can be minimized. The short-circuit current (I_{sc}) indicates that carrier mobility within P3HT is sufficiently high. The stability of I_{sc} at higher ΔT values underscores the robustness of the prepared TEG (Fig. 5).

The power output of the prepared TEG depends on the applied resistance. At low resistance, the voltage drops, the current reaches its maximum, and the power output

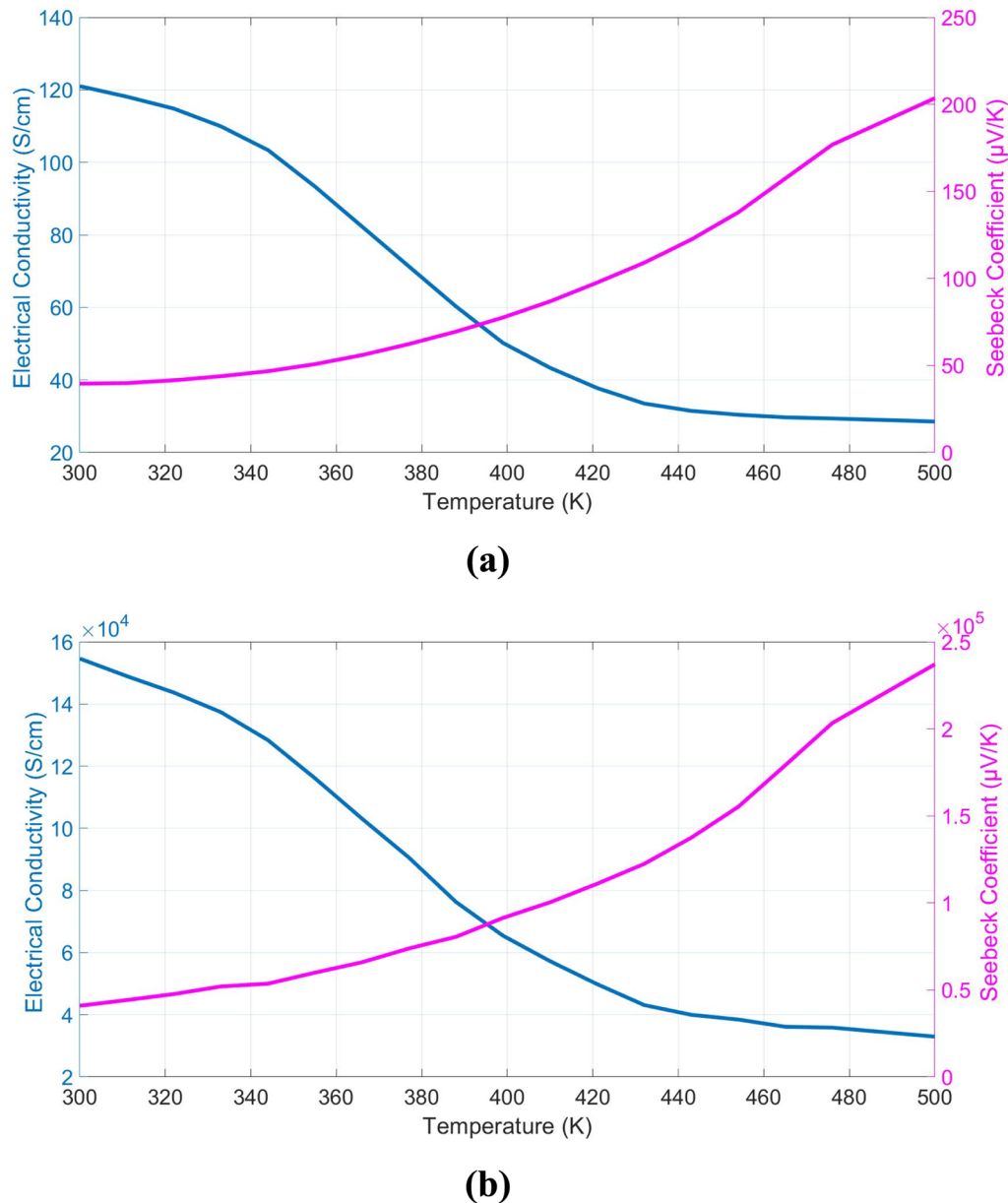


Fig. 4 The electrical conductivity-temperature relation and the Seebeck coefficient-temperature relation of the (a) P3HT-ZnO, and (b) P3HT: Borophene-ZnO P-N junction based OTEGs.

decreases, while at high resistance, the V_{oc} is maintained and the current decreases, reaching the optimum power point (P_{max}). This supports the potential of the prepared TEG for low-grade waste heat recovery applications.

Conclusion

This study successfully demonstrates the enhanced thermoelectric performance of P3HT: Borophene-ZnO p-n junction devices compared to conventional P3HT-ZnO systems. The incorporation of borophene as a conductive

2D additive significantly improves charge transport, yielding a remarkable electrical conductivity of $\sim 12 \times 10^4$ S/cm while maintaining a competitive Seebeck coefficient (100–200 $\mu\text{V/K}$). The synergistic interaction between P3HT, borophene, and ZnO yields an efficient organic-inorganic hybrid structure, particularly effective in the 400–500 K range, where the power factor outperforms that of the binary composite. These findings highlight borophene's critical role in bridging the gap between polymeric and inorganic components, enabling higher carrier mobility and reduced scattering losses. The developed P3HT: Borophene-ZnO OTEGs present a viable route toward

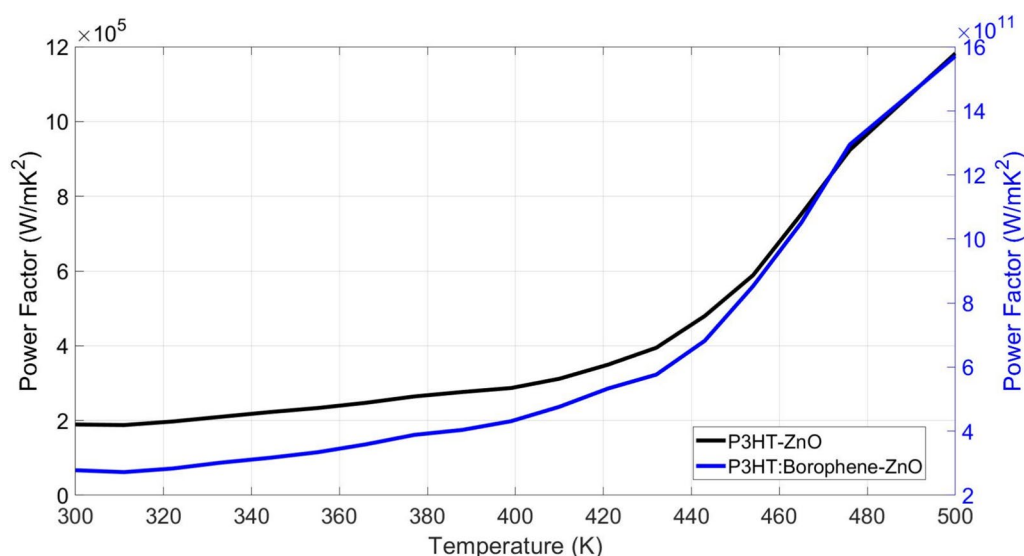


Fig. 5 The power factor-temperature relation of the P3HT:ZnO P-N junction based OTEG, and P3HT: Borophene-ZnO P-N junction based OTEG.

scalable, solution-processable thermoelectric materials for waste heat recovery.

Acknowledgment The authors gratefully acknowledge the Scientific and Technological Research Council of Turkey (TUBITAK) due to the financial support for the Project 122N962.

Author Contributions Ms. AŞ performed the investigation and methodology. Prof. NT performed the investigation, methodology, writing—review & editing, supervision of the study. All authors contributed to the article and approved the final manuscript.

Funding The authors gratefully acknowledge the Scientific and Technological Research Council of Turkey (TUBITAK) due to the financial support for the Project 122N962.

Data Availability The underlying data related to our submission is available on request from nevintasaltin@maltepe.edu.tr.

Competing interest We declare that we have no competing financial, professional, or personal interests that might have influenced the performance or presentation of the work described in this manuscript.

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